

Time Series

Spring Semester 2026, EPFL
École Polytechnique Fédérale de Lausanne
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Practice Exam — Week 13: Linear Time-Invariant (LTI) Filters

Duration: 60 minutes · No calculator · Answer on paper

Name: _____ Surname: _____

SCIPER (Student Card Number): _____

Exercise	Points
1	
2	
3	
4	
Total	

Justify every answer. Throughout, ε_t is a mean-zero white noise of variance σ^2 .

Problem 1 (5 points)

(LTI axioms, impulse response and transfer function.) A digital filter L maps an input sequence to an output sequence, $\{Y_t\} = L[\{X_t\}]$; we write a single output entry as $Y_u = L[\{X_t\}]_u$. Recall that L is *linear time-invariant* (LTI) if it is linear and commutes with the backshift, $LB = BL$. The impulse response is $h_m = L[\{\delta_{t,-m}\}]_0$ and, for $h \in \ell^1$, the transfer function is $H(f) = \sum_{m \in \mathbb{Z}} h_m e^{-2\pi i f m}$.

(a) Let L_1, L_2 be LTI filters and $\alpha \in \mathbb{C}$. Prove, directly from the two axioms, that the linear combination $L = \alpha L_1 + L_2$ is again LTI (check linearity and time invariance separately). (2 pts)

(b) Consider the concrete one-sided filter

$$L[\{X_t\}]_u = X_u - \frac{1}{2} X_{u-1}, \quad u \in \mathbb{Z}.$$

Identify its impulse response $\{h_m\}$, write L as a polynomial $\Phi(B)$ in the backshift, and compute its transfer function $H(f)$ together with the squared gain $|H(f)|^2$ in real form (no complex exponentials). (2

pts)

(c) At which frequency $f \in [0, \frac{1}{2}]$ is the squared gain $|H(f)|^2$ of part (b) *largest*, and at which is it *smallest*? Give both extreme values of $|H(f)|^2$, and state in one sentence whether this filter is low-pass or high-pass. (1 pt)

Problem 2 (5 points)

(Differencing, smoothing, and the output spectrum.) Let $\{X_t\}$ be a zero-mean stationary process with $\gamma^{(X)} \in \ell^1$ and spectral density $\mathcal{S}_X(f)$. Recall the output-spectrum theorem: if $\{Y_t\} = L[\{X_t\}]$ for an LTI filter with transfer function H , then $\mathcal{S}_Y(f) = |H(f)|^2 \mathcal{S}_X(f)$.

(a) Let $\{Y_t\} = (I - B)[\{X_t\}]$, i.e. $Y_t = X_t - X_{t-1}$. Explain briefly why $\{Y_t\}$ is stationary, then prove that

$$\mathcal{S}_Y(f) = 4 \sin^2(\pi f) \mathcal{S}_X(f).$$

(Show the algebraic step $|1 - e^{-2\pi i f}|^2 = 4 \sin^2(\pi f)$.) (2 pts)

(b) Now cascade the difference with the centred three-point average

$$L_2[\{V_t\}]_u = \frac{1}{3}(V_{u-1} + V_u + V_{u+1}),$$

forming $\{W_t\} = L_2[(I - B)[\{X_t\}]]$. Using the fact that the transfer function of a cascade is the *product* of the two transfer functions, express $\mathcal{S}_W(f)$ in terms of $\mathcal{S}_X(f)$, and name every frequency in $[0, \frac{1}{2}]$ at which $\mathcal{S}_W$ is forced to vanish. (2 pts)

(c) Specialise to white-noise input, $\mathcal{S}_X(f) \equiv \sigma^2$, and evaluate $\mathcal{S}_Y(\frac{1}{4})$ for the differenced series of part (a). (1 pt)

Problem 3 (5 points)

(ARMA spectral density and time reversibility of AR(p)). For a stationary ARMA(p, q) process $\Phi(B)X_t = \Theta(B)\varepsilon_t$ the spectral density is

$$\mathcal{S}_X(f) = \sigma^2 \frac{|\Theta(e^{-2\pi i f})|^2}{|\Phi(e^{-2\pi i f})|^2}, \quad \Phi(z) = 1 - \sum_{j=1}^p \phi_j z^j, \quad \Theta(z) = 1 - \sum_{j=1}^q \theta_j z^j.$$

(a) For the AR(1) process $X_t = \phi X_{t-1} + \varepsilon_t$ with $|\phi| < 1$, use the polynomial-filter lemma (the filter $\Phi(B)$ has transfer function $\Phi(e^{-2\pi i f})$) to derive

$$\mathcal{S}_X(f) = \frac{\sigma^2}{1 + \phi^2 - 2\phi \cos(2\pi f)},$$

and state, with reason, where $\mathcal{S}_X$ peaks when $\phi > 0$ and when $\phi < 0$. (2 pts)

(b) Let $\{X_t\}$ be a Gaussian AR(p) process, $X_t = \sum_{j=1}^p \phi_j X_{t-j} + \varepsilon_t$, and define the time-reversed series $Y_t = X_{-t}$. Show that $\gamma_\tau^{(Y)} = \gamma_\tau^{(X)}$ for all τ , hence $\mathcal{S}_Y(f) = \mathcal{S}_X(f)$. (1.5 pts)

(c) Put $\phi_0 = -1$ and $\tilde{Y}_t = \sum_{j=0}^p \phi_j Y_{t-j} = \Phi(B)[\{Y_t\}]_t$. Using parts (a)–(b), the polynomial-filter lemma and the output-spectrum theorem, prove that $\mathcal{S}_{\tilde{Y}}(f) \equiv \sigma^2$, and conclude that $\{Y_t\}$ is an AR(p) with the *same* coefficients $\phi_1, \dots, \phi_p$ driven by white noise of variance σ^2 . (1.5 pts)

Problem 4 (5 points)

(Revision of Week 12: ARCH(1) — moments, white noise, and the ACF of the squares.) Let $\{r_t\}$ be an ARCH(1) process,

$$r_t = \sigma_t \varepsilon_t, \quad \sigma_t^2 = \alpha_0 + \alpha_1 r_{t-1}^2,$$

with $\alpha_0 > 0$, $0 \leq \alpha_1 < 1$, and $\{\varepsilon_t\}$ i.i.d. standard Gaussian (mean 0, variance 1) independent of $\mathcal{F}_{t-1}$.

(a) Using that σ_t is $\mathcal{F}_{t-1}$ -measurable and iterated expectation, show $\mathbb{E}[r_t] = 0$, and (assuming weak stationarity) that

$$\text{Var}(r_t) = \frac{\alpha_0}{1 - \alpha_1}.$$

(1.5 pts)

(b) Prove that $\{r_t\}$ is serially uncorrelated, i.e. $\text{Cov}(r_{t+\tau}, r_t) = 0$ for all $\tau \neq 0$, by conditioning on $\mathcal{F}_{t+\tau-1}$. (This is what makes ARCH a *white noise*.)

(1.5 pts)

(c) Assume in addition $0 < \alpha_1 < 1/\sqrt{3}$ (so a finite fourth moment exists). Recall $\text{Corr}(r_{t+\tau}^2, r_t^2) = \alpha_1^{|\tau|}$ and the kurtosis $\kappa = \frac{3(1 - \alpha_1^2)}{1 - 3\alpha_1^2}$. Take $\alpha_1 = \frac{1}{2}$ and report the numerical value of $\text{Corr}(r_{t+1}^2, r_t^2)$, of $\text{Corr}(r_{t+2}^2, r_t^2)$, and of the kurtosis κ ; state in one sentence why $\kappa > 3$ means fat tails. (2 pts)

